Publishing & Sharing Reports in Power BI

Easy Level

Q1. Explain why publishing and sharing reports is considered more important than just building dashboards in Power BI Desktop.

- **Collaboration & Decision-Making:** Building a report locally on Power BI Desktop keeps data isolated on a single developer's machine. Publishing allows stakeholders and business leaders to access real-time insights from anywhere to make timely decisions.
- **Automated Data Refresh & Centralization:** In Power BI Desktop, data remains static unless refreshed manually by the developer. Publishing to the cloud enables scheduled data refreshes, single-source-of-truth governance, secure access control, and distribution via web, mobile apps, or embedded portals.

Q2. Define Power BI Service and explain its role in comparison to Power BI Desktop in the reporting lifecycle.

- **Power BI Service:** A cloud-based Software as a Service (SaaS) platform used to host, publish, share, collaborate on, and manage Power BI content.
- **Role Comparison in the Lifecycle:**
    - **Power BI Desktop (Authoring Tool):** Used at the start of the lifecycle for data extraction, transformation (Power Query), data modeling, DAX measure creation, and visual design.
    - **Power BI Service (Distribution & Consumption Platform):** Used at the end of the lifecycle for hosting published reports, building high-level dashboards, configuring scheduled refreshes, managing security (RLS), and sharing content with end users.

Q3. What does publishing a report mean in Power BI? List any two outcomes that occur after a report is published.

- **Definition:** Publishing refers to uploading a completed Power BI Desktop file (`.pbix`) from a local computer to a cloud workspace in the Power BI Service.
- **Two Key Outcomes:**
    1. **Dataset & Report Creation:** Two distinct items are created in the target workspace—the report visuals and its underlying semantic model (dataset).

2. **Accessibility & Scheduled Refresh:** The report becomes accessible via web browsers or mobile devices, allowing administrators to configure scheduled data refreshes and share access with authorized users.

Medium Level

Q4. List and explain the steps involved in publishing a report from Power BI Desktop to Power BI Service in the correct sequence.

1. **Save the `.pbix` File:** Ensure all report transformations, measures, and visual formatting are saved locally.
2. **Sign In to Power BI:** Log into your organizational Microsoft account within Power BI Desktop.
3. **Click "Publish":** Navigate to the **Home** tab on the ribbon and click the **Publish** button.
4. **Select Target Workspace:** Choose the destination Power BI Service workspace (e.g., *My Workspace* or a shared Team/Department workspace).
5. **Publish and Confirm:** Click **Select**. Power BI Desktop uploads the dataset and report. Once complete, a success dialog provides a direct web link to open the report in Power BI Service.

Q5. Differentiate between Report and Dashboard in Power BI based on structure and usage.

Feature	Report	Dashboard
Origin / Source	Created in Power BI Desktop or Service; built from a **single dataset**.	Created **only in Power BI Service**; created by pinning visuals (tiles) from **multiple reports/datasets**.
Structure	Multi-page, interactive canvases with slicers, cross-highlighting, and drill-downs.	Single-page high-level view (canvas) displaying summary key metrics and tiles.
Usage	Detailed, slice-and-dice interactive analysis for operational users and analysts.	Executive-level monitoring of high-level KPIs and metrics across various business units.

Q6. Explain the difference between the following three sharing methods in Power BI Service and mention one use case for each:

1. **Direct Report Sharing:**
    - *Explanation:* Sharing an individual report directly with specific users via email or generated link.

    - *Use Case:* Sharing a quick weekly sales report with an individual account manager for immediate review.
2. **Workspace Access:**
    - *Explanation:* Granting a user access to the entire workspace where all reports, datasets, and dashboards reside.
    - *Use Case:* Giving team members (co-developers or data analysts) access so they can collaborate on building and editing reports.
3. **Power BI Apps:**
    - *Explanation:* Packaging a collection of related dashboards and reports into a single, polished, read-only application for distribution.
    - *Use Case:* Distributing an official, production-ready Sales & Marketing KPI Portal to hundreds of company employees across an enterprise.

Q7. Describe the four workspace roles available in Power BI and explain how access control is managed using these roles.

- **Admin:** Full control over the workspace; can manage workspace settings, delete the workspace, assign/change workspace roles, and edit/publish content.
- **Member:** Can add/modify content within the workspace, reshare items, and publish/update Apps, but cannot delete the workspace or modify Admin permissions.
- **Contributor:** Can create, edit, and delete reports and datasets within the workspace, but cannot publish Apps or modify workspace permissions.
- **Viewer:** Read-only access to view and interact with reports and dashboards; cannot edit content, view underlying datasets, or modify settings.
- **How Access Control is Managed:** Workspace admins use these roles to enforce the **Principle of Least Privilege**—reserving Admin/Member roles for core developers, Contributor roles for content creators, and Viewer roles for end users to prevent accidental deletion or corruption of assets.

Hard Level

Q8. Why are Power BI Apps considered more secure and scalable than direct sharing? Explain this from a governance and enterprise usage perspective.

- **Governance & Staging Environment:** Workspace access acts as a staging/development area. Apps decouple the development environment from the end-user environment. Developers can freely update reports in the workspace without affecting what business users see until the App is explicitly updated and published.
- **Permission Management & Scalability:** Direct sharing with individual users leads to permission sprawl, making auditing difficult. Apps allow centralized permission management through Microsoft Entra ID (Azure AD) security groups, enabling access management for thousands of users simultaneously.

- **Preventing Data Corruption:** Apps restrict end-users to read-only interactions, preventing unauthorized users from editing visuals, breaking data models, or exposing underlying datasets.

Q9. Explain Row-Level Security (RLS) in Power BI. How does RLS ensure that different users see different data using the same report?

- **Definition:** Row-Level Security (RLS) is a security feature used to restrict data access at the row level for specific users based on rules/roles defined within the semantic model using DAX filters.
- **How It Works:**
    1. Developers define roles in Power BI Desktop using DAX filter expressions (e.g., `[Region] = USERPRINCIPALNAME()`).
    2. Users are assigned to these roles in Power BI Service.
    3. When a user opens the report, Power BI dynamically evaluates their login identity (`USERPRINCIPALNAME()`) and filters the dataset at the database level before rendering visuals. Thus, a Regional Sales Manager for *North* and a Regional Sales Manager for *South* open the exact same report, but each only sees rows corresponding to their assigned region.

Q10. Assume an organization shares reports without using Apps, workspace roles, or RLS. Discuss any three risks or issues that may arise related to security, governance, or data misuse, and explain how Power BI best practices help prevent them.

1. **Unauthorized Data Exposure & Leakage:**
    - *Risk:* Without RLS or controlled sharing, sensitive data (such as salary figures, regional profits, or personal client details) becomes visible to all users who open the report.
    - *Prevention:* Implement **Row-Level Security (RLS)** to filter rows dynamically based on user role and identity.
2. **Version Control Chaos & Accidental Overwrites:**
    - *Risk:* Without assigned Workspace Roles (e.g., allowing everyone edit access), users might accidentally overwrite measures, delete pages, or publish untested changes directly into production view.
    - *Prevention:* Restrict workspace creation to core teams, assign end-users the **Viewer** role, and publish content through **Power BI Apps** to keep staging separate from production.
3. **Permission Sprawl & Audit Failures:**
    - *Risk:* Relying solely on ad-hoc direct link sharing creates unmanageable permission lists, making it impossible for IT auditors to track who has access to company data or revoke access when employees leave.

- *Prevention:* Manage distribution at scale using **Power BI Apps** mapped to managed **Entra ID Security Groups**.